

A critical L^1 counterexample for the harmonic Fock projection

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Abstract

Vujadinovic conjectures that the harmonic Fock projection P_α is bounded on $L^p(\mathbb{R}^n, d\mu_\beta)$, for $p \geq 1$, if and only if $p\beta = 2\alpha$, and claims this theorem when n is even. We give an elementary counterexample at the critical endpoint. In dimension four, P_α is not bounded on $L^1(\mathbb{R}^4, d\mu_{2\alpha})$ for any $\alpha > 0$. The exact four-dimensional kernel turns a column centered at Re_1 into a fixed Gaussian times an oscillatory expression of size R . Its absolute mass is bounded below by $cR - C$. Since bounded L^1 integral operators have uniformly bounded absolute columns, the result follows. This disproves Conjecture 1.6(b) and also contradicts the source's claimed even-dimensional endpoint theorem.

1 Exact source question and definitions

For $\gamma > 0$, let

$$d\mu_\gamma(x) = (\pi\gamma)^{-n/2} e^{-|x|^2/\gamma} dx.$$

The harmonic Fock space $\mathcal{H}_\alpha^2$ consists of the harmonic functions in $L^2(\mathbb{R}^n, d\mu_\alpha)$, and its orthogonal projection is

$$P_\alpha f(x) = \int_{\mathbb{R}^n} H_\alpha(x, y) f(y) d\mu_\alpha(y).$$

The source states:

Conjecture 1.6. Suppose $\alpha > 0$, $\beta > 0$ and $0 < p < \infty$. Then: (a) P_α is not bounded on $L^p(\mathbb{R}^n, d\mu_\beta)$ for $0 < p < 1$; (b) for $p \geq 1$, P_α is bounded on $L^p(\mathbb{R}^n, d\mu_\beta)$ if and only if $p\beta = 2\alpha$.

The paper's Theorem 1.4(b) claims part (b) for every even n . The exact source page is reproduced below.

Authors in [8] observed n -dimensional complex space $\mathbb{C}^n$ with the Gaussian probability measure $dv_t(z) = \left(\frac{t}{\pi}\right)^n e^{-t|z|^2} dv(z)$, $t > 0$, where dv is Lebesgue measure on $\mathbb{C}^n$. The analytic Fock space F_t^p consists from all entire functions in $L^p(\mathbb{C}^n, dv_t)$, $p > 0$. The reproducing kernel associated to F_t^2 is $K_t(x, y) = e^{t\langle x, y \rangle}$.

The operators under the consideration in [8] are

$$S_t f(z) = \int_{\mathbb{C}^n} e^{t\langle z, w \rangle} f(w) dv_t(w)$$

and

$$T_t f(z) = \int_{\mathbb{C}^n} |e^{t\langle z, w \rangle}| f(w) dv_t(w),$$

$f \in L^p(\mathbb{C}^n, dv_t)$.

The main result of [8] deals with the problem of boundedness of operators S_t and T_t on $L^p(\mathbb{C}^n, dv_s)$ for fixed parameters $s > 0, t > 0$. In fact, it is proved that T_t (S_t) is bounded on $L^p(\mathbb{C}^n, dv_s)$ if and only if $pt = 2s$.

A natural and important question which arises in the context of harmonic Fock spaces is to determine the conditions under which the operator P_α is bounded on $L^p(\mathbb{R}^n, d\mu_\beta)$ for various choices of parameter $\beta > 0$.

The following theorem presents the main result of this paper and it gives an answer to the previously raised question under certain conditions.

Theorem 1.4. *Suppose $\alpha > 0, \beta > 0$ and $0 < p < \infty$. Then:*

- a) *The operator P_α is not bounded on $L^p(\mathbb{R}^n, d\mu_\beta)$ for $0 < p < 1$.*
- b) *For $p \geq 1$ and $n = 2N, N \geq 1$, the operator P_α is bounded on $L^p(\mathbb{R}^n, d\mu_\beta)$ if and only if*

$$p\beta = 2\alpha.$$

As the matter of fact, we prove that $p\beta = 2\alpha$ is a necessary condition for the boundedness of the operator P_α for any $n \geq 2$.

As an immediate consequence we obtain the following result.

Corollary 1.5. *Let $n = 2N, N \geq 1$ and $\alpha > 0$. For $p \geq 1$ the operator P_α is a bounded projection from L_α^p onto $\mathcal{H}_\alpha^p$.*

It remains an open problem to extend the Theorem 1.4 for the case of an arbitrary dimension $n \geq 3$. It seems that the main problem is produced from the fact that the formula (1.16) provides an explicit formula (1.17) only for even number n . However, we conjecture the sequel extended result.

Conjecture 1.6. *Suppose $\alpha > 0, \beta > 0$ and $0 < p < \infty$. Then:*

- a) *The operator P_α is not bounded on $L^p(\mathbb{R}^n, d\mu_\beta)$ for $0 < p < 1$.*
- b) *For $p \geq 1$ the operator P_α is bounded on $L^p(\mathbb{R}^n, d\mu_\beta) \Leftrightarrow p\beta = 2\alpha$.*

In the sequel, the dimension n is considered to be the integer such that $n \geq 2$, unless stated otherwise.

We disprove the sufficiency direction already for $n = 4, p = 1$ and $\beta = 2\alpha$.

2 The exact four-dimensional kernel

It is enough to take $\alpha = 1$. Indeed, the changes of variables $x = \sqrt{\alpha}X$ and $y = \sqrt{\alpha}Y$ carry $d\mu_\alpha$ to $d\mu_1, d\mu_{2\alpha}$ to $d\mu_2$, and

$$H_\alpha(\sqrt{\alpha}X, \sqrt{\alpha}Y) = H_1(X, Y).$$

For completeness, we derive the needed kernel directly from the zonal series quoted in the source. Put

$$s = |x||y|, \quad \cos \theta = \frac{\langle x, y \rangle}{|x||y|}.$$

In dimension four, the normalized zonal harmonic of degree k is $(k+1)C_k^1(\cos \theta)$ and $(2)_k = (k+1)!$. Consequently

$$\begin{aligned} H_1(x, y) &= \sum_{k=0}^{\infty} C_k^1(\cos \theta) \frac{s^k}{k!} \\ &= \frac{1}{\sin \theta} \operatorname{Im} \left(e^{i\theta} e^{se^{i\theta}} \right). \end{aligned}$$

Writing

$$A = \langle x, y \rangle = s \cos \theta, \quad B = \sqrt{|x|^2|y|^2 - \langle x, y \rangle^2} = s \sin \theta,$$

we obtain the exact formula

$$\boxed{H_1(x, y) = e^A \left(A \frac{\sin B}{B} + \cos B \right)}, \quad (1)$$

where $\sin B/B$ is interpreted as 1 at $B = 0$. This also agrees with Englis's independent closed kernel formula [2, Corollary 7].

3 Absolute columns and the counterexample

Relative to $d\mu_2$, the same projection is

$$P_1 f(x) = \int_{\mathbb{R}^4} K(x, y) f(y) d\mu_2(y), \quad K(x, y) = 4e^{-|y|^2/2} H_1(x, y), \quad (2)$$

because $d\mu_1(y) = 4e^{-|y|^2/2} d\mu_2(y)$.

We first record the elementary endpoint fact that will turn an exact column calculation into operator unboundedness.

Lemma 1 (bounded L^1 kernels have bounded columns). *Let ν be a locally finite measure with full support on $\mathbb{R}^d$, and let K be continuous. If*

$$Tf(x) = \int K(x, y) f(y) d\nu(y)$$

extends boundedly on $L^1(\nu)$, then, for every y ,

$$\int |K(x, y)| d\nu(x) \leq \|T\|_{L^1 \rightarrow L^1}.$$

Proof. Let f_ε be the indicator of $B(y, \varepsilon)$ divided by its ν -measure. Then $\|f_\varepsilon\|_1 = 1$ and, by continuity, $Tf_\varepsilon(x) \rightarrow K(x, y)$ for every x . Fatou's lemma gives

$$\int |K(x, y)| d\nu(x) \leq \liminf_{\varepsilon \downarrow 0} \|Tf_\varepsilon\|_1 \leq \|T\|.$$

□

Theorem 1. *For every $\alpha > 0$, the harmonic Fock projection P_α is not bounded on $L^1(\mathbb{R}^4, d\mu_{2\alpha})$.*

Proof. By scaling, take $\alpha = 1$. For $R > 0$, set $y_R = Re_1$ and parameterize

$$x = (R + u, v), \quad u \in \mathbb{R}, \quad v \in \mathbb{R}^3, \quad \text{quad} \rho = |v|.$$

Then $A = R(R + u)$ and $B = R\rho$. Combining (1), (2), and $d\mu_2(x) = (2\pi)^{-2}e^{-|x|^2/2}dx$ gives the exact absolute column mass

$$\begin{aligned} J(R) &:= \int_{\mathbb{R}^4} |K(x, y_R)| d\mu_2(x) \\ &= \frac{1}{\pi^2} \int_{\mathbb{R}} \int_{\mathbb{R}^3} e^{-(u^2 + |v|^2)/2} \left| (R + u) \frac{\sin(R|v|)}{|v|} + \cos(R|v|) \right| dv du. \end{aligned} \quad (3)$$

The decisive cancellation in the Gaussian exponent is exact:

$$-\frac{R^2}{2} - \frac{(R + u)^2 + |v|^2}{2} + R(R + u) = -\frac{u^2 + |v|^2}{2}.$$

Restrict (3) to $|u| \leq 1$ and $1 \leq \rho \leq 2$. For $R \geq 2$, reverse triangle inequality gives

$$\left| (R + u) \frac{\sin(R\rho)}{\rho} + \cos(R\rho) \right| \geq \frac{R - 1}{2} |\sin(R\rho)| - 1.$$

Furthermore,

$$\begin{aligned} \int_1^2 |\sin(R\rho)| d\rho &= \frac{1}{R} \int_R^{2R} |\sin t| dt \\ &\geq \frac{2}{R} \left\lfloor \frac{R}{\pi} \right\rfloor \geq \frac{2}{\pi} - \frac{2}{R}. \end{aligned}$$

On $1 \leq \rho \leq 2$, the radial Gaussian density $e^{-\rho^2/2}\rho^2$ is at least e^{-2} . Passing to polar coordinates in v therefore yields constants $c, C > 0$, independent of R , such that

$$J(R) \geq cR - C \longrightarrow \infty.$$

Lemma 1, applied to the continuous kernel (2), contradicts boundedness on $L^1(d\mu_2)$. Scaling back proves the assertion for every $\alpha > 0$. $\square$

Remark 1 (explicit test functions). *The proof is constructive. For fixed R , let*

$$f_{R,\varepsilon} = \frac{\mathbf{1}_{B(y_R, \varepsilon)}}{\mu_2(B(y_R, \varepsilon))}.$$

Then $\|f_{R,\varepsilon}\|_{L^1(d\mu_2)} = 1$ and Fatou gives

$$\liminf_{\varepsilon \downarrow 0} \|P_1 f_{R,\varepsilon}\|_{L^1(d\mu_2)} \geq J(R).$$

Choosing first $R \rightarrow \infty$ and then $\varepsilon \downarrow 0$ produces an explicit norm-divergent family.

4 What fails in the source proof

The source’s Theorem 2.7 claims boundedness at $p = 1$, $\beta = 2\alpha$ by estimating the absolute kernel integral

$$I_\alpha^\beta(y) = \int |H_\alpha(y, x)| d\mu_\beta(x).$$

In the cone calculation leading to its equations (2.24)–(2.25), a difference of exponentials is replaced by a mean-value expression involving a parameter

$$\theta \in (\sqrt{1 - \varepsilon^2}, 1).$$

The subsequent Kummer estimate treats this as uniform exponential slack strictly below 1. But the mean-value parameter depends on the radial variables and on $|y|$ and can approach 1. Formula (3) shows concretely that no uniform slack exists: on a fixed Gaussian tube around y_R , the absolute mass is of order R .

This counterexample directly contradicts Theorem 1.4(b), Corollary 1.5, and Theorem 2.7 of the source at $n = 4$, $p = 1$. It disproves Conjecture 1.6(b) as written. It does *not* decide whether the critical projection could still be bounded for $1 < p < \infty$ in odd dimensions; cancellation in the input variable can matter away from the L^1 endpoint.

5 Verification and novelty check

The accompanying script `verify_column_growth.py` performs four independent checks:

1. it compares (1) against high-order partial sums of the zonal-harmonic series;
2. it checks the exact Gaussian exponent cancellation;
3. it quadratures (3) for increasing R ;
4. it compares the numerical columns with the explicit restricted-region lower bound used above.

Targeted searches for the exact title, Conjecture 1.6, Theorem 2.7, critical L^1 counterexamples, corrections, and errata found the source’s 2022 journal publication but no correction or later answer. Before public use, a specialist should independently audit the kernel normalization and Lemma 1, search MathSciNet, zbMATH, and citation databases, and notify the source author and journal if the result is confirmed.

References

- [1] D. Vujadinovic, *Boundedness of the orthogonal projection on Harmonic Fock spaces*, Complex Anal. Oper. Theory **16** (2022), article 13; arXiv:1902.08417.
- [2] M. Engliš, *Berezin transform on the harmonic Fock space*, J. Math. Anal. Appl. **367** (2010), 75–97.